

Analysis of Performance Drop After Copy-Paste Augmentation

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April 27, 2026

1 Brief Summary

This report explains why the copy-paste augmented datasets reduce performance. The short answer is simple: the new samples do not clearly align with the test distribution, and in Global Wheat Head the score-guided version is even farther from test than random copy-paste.

2 Research Problem

The main question is whether the augmented images remain valid training samples or become distribution-shifted samples that hurt generalization. This is the problem I am investigating now because performance drops after adding copy-paste augmentation.

To study that, I used a frozen YOLOv8s backbone, extracted image-level embeddings from the P5 feature map with global average pooling, and compared original training images, random augmented images, score-guided augmented images, and test images.

3 Scoring

The scoring system is the bridge between the baseline model and augmentation. It uses the model's own failure behavior to rank difficult objects and images. Low-confidence predictions measure object difficulty, and optimal-threshold predictions measure false positives. The idea is to select hard samples more often. The problem is that harder samples are not automatically closer to the test distribution.

4 Visual Comparison

Figure 1 shows the same style of augmented-vs-base comparison used in the notebook viewer. It is the clearest visual check for whether the new image still looks like a plausible sample from the original dataset.

5 Feature-Space Analysis

This is the key result.

5.1 MinneApple

MinneApple uses same-image copy-paste. The feature-space plot shows that original train, random augmentation, and score-guided augmentation stay close to each other, while test is separated. So the augmented images are not obvious outliers, but they also do not move toward the test distribution. The mechanism is still not fully clear.

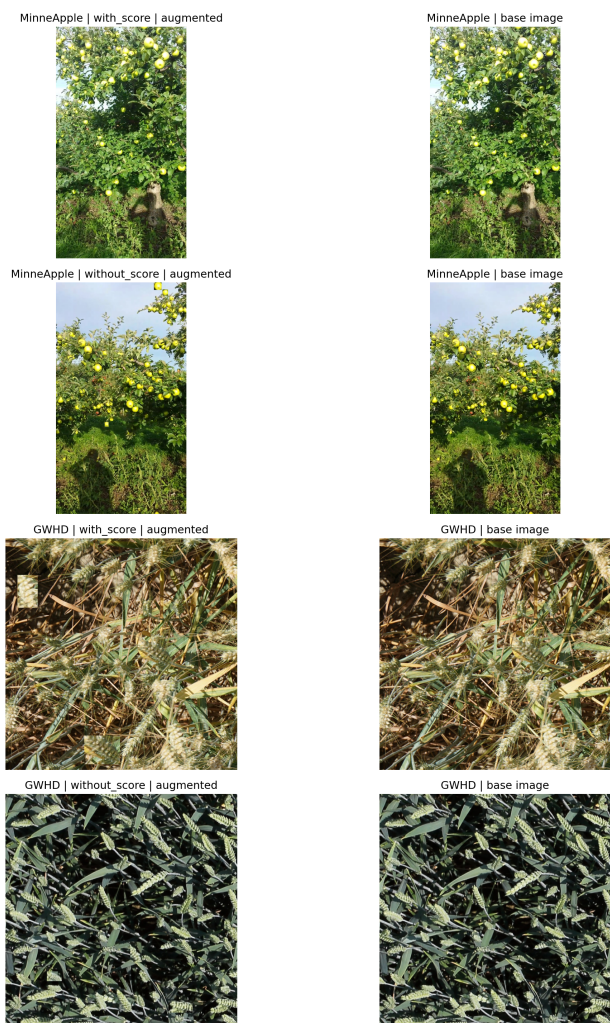


Figure 1: Representative augmented-vs-base comparison for MinneApple and Global Wheat Head.

5.2 Global Wheat Head

Global Wheat Head is more problematic because cross-image copy-paste mixes domains. The centroid distances are:

- original train $\rightarrow$ test: 16.8058
- augmented random $\rightarrow$ test: 16.2602
- augmented score $\rightarrow$ test: 17.3665

The sharp observation is that score-guided augmentation is the farthest from test. That matches the performance drop. What is still not clear is the exact causal mechanism: whether the drop is caused by density, domain mixing, poor placement, or the scoring rule itself.

6 Conclusion

The conclusion is direct: the current copy-paste augmentation does not reliably improve the training distribution, and score-guided selection is not the fix. For MinneApple, the samples stay close to the training manifold but still do not improve test alignment. For Global Wheat Head, score-guided augmentation is clearly the most misaligned with test. So the performance drop is consistent with distribution mismatch, but the exact reason is still not fully resolved.

7 Detailed Experimental Setup

This final part records how the experiments were made and what they produced.

7.1 Experiment sources

- MinneApple result root: `results/12_multiseed_score_vs_noscore_copy_paste_exp_apple_withaug`
- Global Wheat Head result root: `results/13_multiseed_score_vs_noscore_copy_paste_exp_global_whe`
- Feature analysis notebooks: `notebooks/augmentation_feature_space_outlier_analysis.ipynb` and `notebooks/augmentation_feature_space_outlier_analysis_gwhd.ipynb`

7.2 Configuration table

Setting	MinneApple	Global Wheat Head
Backbone	YOLOv8s	YOLOv8s
Feature level	P5	P5
Embedding method	GAP over feature map	GAP over feature map
Seed count	1 for feature analysis, 3 for comparison	3
Copy-paste mode	same-image-only for MinneApple	cross-image for GWHD
Augmentation variants	random vs score-guided	random vs score-guided
Comparison target	test distribution distance	test distribution distance

7.3 Result table: Global Wheat Head

Condition	COCO AP	AP50	AP75
Baseline mean	0.50699	0.92226	0.49071
With-score mean	0.48367	0.90454	0.45494
Without-score mean	0.48570	0.90476	0.45556
With-score Δ vs baseline	-0.02332	-0.01773	-0.03578
Without-score Δ vs baseline	-0.02129	-0.01750	-0.03515

7.4 Result table: MinneApple

Condition	COCO_AP	COCO_AP50	COCO_AP75	AP_small	AP_medium	AP_large
Baseline	0.4472	0.7743	0.4637	0.3075	0.5996	0.7191
Random copy-paste	0.4362	0.7695	0.4532	0.2903	0.5933	0.7158
Score-guided copy-paste	0.4280	0.7882	0.4288	0.2979	0.5748	0.6271

7.5 Observation from the tables

On both datasets, the augmented results are below the baseline on COCO_AP. On Global Wheat Head, the score-guided variant is consistently worse than random. On MinneApple, score-guided copy-paste gives the weakest AP75. That is enough to support the main claim, but not enough to explain every detail of the failure.

8 Appendix

The notebook outputs and the raw experiment summaries used for this report are stored under the corresponding **results/** directories.